

# Intrinsic Resting-State Connectivity and Personal Meaning Across the Transition to Fatherhood

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New fatherhood is marked by profound psychological and neural transformations. While these changes can be stressful, many fathers also report increased life satisfaction associated with parenthood. Personal meaning—the subjective sense that life has purpose, meaning, and coherence—may reflect positive adaptation to parenthood. This study investigated how personal meaning across the transition to fatherhood is associated with both psychological and neural changes. Among 88 first-time fathers followed from mid-pregnancy into the first year after birth, we found an approximately even split between fathers who reported increases or decreases in personal meaning. Within 35 fathers who also underwent neuroimaging before and after birth, we observed increases in resting-state functional connectivity in the bilateral temporal lobes, right angular gyrus, thalamus, and right lateral occipital cortex, and decreases in regions such as the right frontal pole, left opercular cortex, and anterior cingulate. Increases in insular cortex connectivity predicted greater postnatal personal meaning, even after controlling for both positive and negative parenting-related feelings. These findings suggest that dynamic changes within the insula may support fathers' ability to form a higher order sense of meaning and purpose during this transformative period. Future mixed-methods research may further elucidate how these neural changes contribute to well-being during stressful life transitions.

**Keywords:** meaning and purpose, fatherhood, transition to parenthood, resting-state connectivity, temporal pole

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The transition to fatherhood brings psychosocial changes across many domains, including new social roles and relationships. Fatherhood can be broadly defined as a multifaceted role encompassing biological, psychological, and social dimensions, with responsibilities in child-rearing and the broader family system that vary across individuals and cultural contexts. This period may also bring an increased risk of psychological distress related to the significant and sudden life changes. Although many studies have focused on negative experiences of parenthood, most new parents experience *mixed feelings* in early parenthood, characterized by positive and negative sentiment toward the same situation (Baldwin et al., 2018; Blakey et al., 2025; Fridbäck et al., 2024; Meleagrou-Hitchens & Willig, 2022; Vaccaro et al., 2020). How parents derive meaning from this complex transition may reflect their well-being and adaptation to parenthood. Additionally, neural changes during

this transformative period may provide a window into how brain and behavior interact to create beneficial or harmful change (Kim, 2016; Saxbe et al., 2018).

## Meaning in the Transition to Parenthood

Personal meaning is a subjective sense that one's life is purposeful and coherent, with a clear sense of direction and goals (Reker, 2005; Reker & Peacock, 1981). This construct captures how individuals integrate their life experiences into a broader sense of identity, purpose, and agency, regardless of whether their day-to-day experiences are positive or negative. Personal meaning is particularly important during times of identity and role change, such as new parenthood, as a coherent sense of purpose and identity may provide emotional stability and support resilience. This sense of meaning has

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been found to be a protective factor for mental health during stressful life changes (Boreham & Schutte, 2023; Dulaney et al., 2018; Krause, 2007; Lin, 2021; Park & Baumeister, 2017). A large-scale study using the Midlife in the United States (MIDUS) data set found that a sense of meaning buffered the negative effects of stressful days on well-being, leading to less negative affect, even when positive affect itself was not increased (Hill et al., 2018).

The role of meaning may become particularly salient in the context of parenting, where the long-term benefits of life purpose and coherence may buffer the impact of everyday stressors and challenges on psychological well-being. Many of the stressors related to new parenthood involve responsibilities that fundamentally alter one's daily life, such as the intensive demands of infant care and shifting roles within the family (Abidin, 1983; Saxbe et al., 2018). Previous research suggests that the transition to fatherhood involves shifts in identity and how one finds meaning in life (Genesoni & Tallandini, 2009; Höfner et al., 2011; Rilling & Hadley, 2023; Strauss & Goldberg, 1999). Parents also often report higher levels of meaning than nonparents, further demonstrating the relevance of developing meaning during the transition to parenthood (Nelson et al., 2013; Senior, 2014). However, other studies have found that meaning does not necessarily increase in parents and that factors such as lack of support, changing responsibilities and life limitations, and unmet expectations about the experience of parenthood may complicate the relationship between personal meaning and becoming a parent (Nelson et al., 2014; Pancer et al., 2000). For fathers, questions of identity and meaning may be particularly fraught given that men's role in parenting has evolved in the past half-century, with increased societal expectations for fathers to be hands-on caregivers even as they are also expected to be successful in workplace roles. Our sample reflected men who were cohabiting with their pregnant partners and expected to be involved in childrearing after birth, but we expected to see variability in meaning and parenting stress given fathers' different levels of engagement in parenting.

### Personal Meaning and the Brain

Although there is limited neuroscience research on meaning in life, the studies that have been done suggest important roles for neural pathways involved in integrating emotion, sense of self, and conceptual associations. While there is not enough prior work to justify a region-of-interest approach, previous studies give us reason to hypothesize that insular connectivity and default-mode network connectivity may be particularly important. A recent study by Mwilambwe-Tshilobo et al. (2019) found that in a large sample from the Human Connectome Project, meaning in life was significantly associated with resting-state connectivity between default mode and limbic networks in the positive direction. Functional connections between these networks may facilitate the integration of affective aspects of the environment with a sense of self and autobiographical memory, facilitating meaning making throughout emotionally complex times. These findings align with evidence that parenting involves heightened emotional and cognitive demands, further emphasizing the brain's role in integrating diverse experiences into a coherent sense of purpose.

A cross-sectional study of 70 college students found that right insular cortex gray matter volume was positively associated with self-reported purpose in life, controlling for age and gender (Lewis et al., 2014). A more recent study leveraging a large data set of

resting-state functional magnetic resonance imaging scans ( $N = 970$ ) found right insular connectivity to the rest of the brain predicted meaning in life measures via the National Institute of Health Toolbox's Meaning and Purpose survey (Zhang et al., 2023). The right insular cortex's connectivity with the rest of the brain, as well as its gray matter density, have been found to significantly predict self-reported meaning in life, further highlighting the relevance of affective processing to meaning making (Lewis et al., 2014; Zhang et al., 2023). Other work has also found that regions of the default-mode network (DMN), which broadly has similar involvements in self-referential processes, tends to associate with higher meaning in life as opposed to day-to-day affect. The gray matter volume of the precuneus, part of the DMN, has been found to positively associate with purpose in life (Sato et al., 2015). Longer term changes in these self-referential networks may be especially important to personal development (Feng et al., 2018; Pfeifer & Peake, 2012). These regions have also been identified as susceptible to change during the transition to parenthood more generally (Paternina-Die et al., 2024), suggesting that part of this neurodevelopment may be related to building a coherent sense of meaning around new life circumstances.

### Brain Changes in Parenthood

A few studies of mothers have reported robust decreases in gray matter volume from before to after birth (Hoekzema et al., 2017, 2022; Paternina-Die et al., 2024; Pritschet et al., 2024; Servin-Barthet et al., 2023). These reductions have been associated with mothers' reports of their preparation for birth and subsequent bonding with their infant (Hoekzema et al., 2017, 2022). Although there is more limited research on fathers, an international collaboration including 20 fathers from Hoekzema et al.'s (2017) study and 20 fathers scanned by our group also found cortical volume reductions in men from pre- to postnatal (Martínez-García et al., 2023). In a follow-up study of 38 fathers who are also included in the present study, we found that these gray matter reductions were associated with bonding to the infant and time devoted to parenting but also with postnatal reports of depression, anxiety, psychological distress, and sleep problems (Saxbe & Martínez-García, 2024).

These structural changes may also support adaptive functional changes that occur during the transition to fatherhood. A study by Diaz-Rojas et al. comparing expectant fathers and childless men found gray matter reductions in the insular cortex (2021). Additionally, the study found that insular activity in response to infant stimuli could significantly classify whether the men were expectant fathers or other childless men using multivariate pattern analysis, suggesting that the overall changes to the insula were related functional adaptations. There are very few studies on functional connectivity changes that accompany the transition to parenthood. For example, studies in mothers have identified perinatal increases in connectivity in key limbic structures such as between the amygdala and the caudate and nucleus accumbens (Dufford et al., 2019). Similarly, increases in DMN connectivity during the postnatal period have been associated with caregiving behavior and bonding (Hoekzema et al., 2022; Orchard et al., 2023). In fathers, findings are even more limited. One study found no significant global changes in functional connectivity between expectant and new fathers but noted that increased amygdala connectivity to the broader mentalizing network (specifically parietal regions) correlated with the time fathers spent in caregiving roles (Horstman et al., 2022).

Despite its small size, the emerging literature on the parenting brain suggests that neural adaptations during parenthood may support sensitive caregiving behaviors while simultaneously increasing susceptibility to psychological distress. These functional changes in the brain highlight the dual nature of parenting as both a period of opportunity for connection and a time of increased vulnerability. However, these neural changes do not occur in isolation—they are deeply intertwined with psychological factors such as how parents derive personal meaning from their experience of parenthood.

## The Present Study

This study aimed to investigate how first-time fathers' reports of personal meaning change during the transition to fatherhood, and how this change relates to feelings surrounding parenting and to resting-state connectivity across the transition to parenthood. To our knowledge, this is the first study to investigate whole-brain resting-state connectivity changes during the transition to fatherhood, how meaning relates to parenting experiences across this transition, and the first to look longitudinally at how neural changes during a major life transition relate to a sense of meaning. We tested four research questions. First, we investigated whether personal meaning changed or showed stability across the transition to first-time fatherhood, predicting that most fathers would report greater personal meaning after birth. Next, we investigated whether fathers' sense of meaning was related to their parenting stress, predicting that meaning would be negatively associated with parenting stress. Third, we examined how resting-state functional connectivity changed during the transition to fatherhood. Based on previous work on the parenting brain and on meaning, we hypothesized that we would primarily see changes in the connectivity of the insula and in DMN regions. Finally, we tested whether changes in resting-state intrinsic functional connectivity would be related to men's sense of personal meaning after birth, hypothesizing that both postnatal connectivity, and the change from prenatal to postnatal connectivity, of the DMN and insular cortex would be positively correlated with postnatal personal meaning, reflecting successful meaning making during the complex transition to parenthood. We hypothesized these associations would hold even when controlling prenatal levels of meaning.

We additionally performed a follow-up exploratory analysis to test whether associations between fathers' sense of meaning and their brain connectivity are distinct from those related to positive and negative dimensions of parenting—thereby exploring whether brain changes are still associated with meaning making when considering the range of positive and negative feelings involved in early parenting.

## Method

### Participants

All research protocols were approved by the University of Southern California's Institutional Review Board. Fathers in the study were drawn from a larger longitudinal study following 100 cohabiting heterosexual couples in the greater Los Angeles area who were expecting their first child. Participants were recruited through flyers and social media advertising. All participating couples met the following eligibility criteria: (a) mixed-sex couples who were currently cohabiting and planned to continue cohabiting after their child's birth;

(b) first pregnancy for both expectant parents; (c) viable singleton fetus at the time of participation; and (d) parents were not taking psychotropic medications that might interfere with hormone or brain scanning. Eligible couples were initially invited to the lab during mid-to-late pregnancy where they provided informed consent. For this study, we dropped any participants who did not complete all of the questions of the Life Attitudes Profile. The present study focused only on the 88 fathers who provided self-reported data on personal meaning both prenatally and again at 6 months after birth.

### Functional Magnetic Resonance Imaging Participants

A subsample of the fathers who participated in the broader study also underwent magnetic resonance imaging. All fathers had no history of neurological illness. Forty-two first-time fathers were scanned during their partners' pregnancy and, of those, 38 fathers returned for a second scan after the postnatal visit. The second scan was originally designed to take place between 6 and 9 months after birth, but for the last 10 fathers in the sample, the timing of postnatal scan visits was affected by COVID-19 lockdowns. Five of these 38 fathers completed the scan over 12 months after their child's birth due to pandemic-related delays. Due to this significant delay, we controlled for the time between the 6-month survey and the postnatal scans in all analyses. For fathers scanned before March 2020 ( $n = 28$ ), the average baby age at the time of the second scan was 30.28 weeks (6.96 months, range = 25.14–48.29 weeks). For those scheduled during or after March 2020 ( $n = 10$ ), the average was 59 weeks (13.57 months, range = 34.28–88.57 weeks). Overall, the average interval was 38 weeks (8.7 months) with a median of 31 weeks (7 months).

Three subjects were excluded from the resting-state studies due to scanning artifacts, leading to a total sample size of 35 in the magnetic resonance imaging substudy (Table 1).

### Questionnaire Measures

Self-report questionnaires were administered during the partner's pregnancy and again at the laboratory visit that occurred approximately 6 months after birth. Prenatal measurements were collected when partners were approximately 20–36 weeks ( $M = 199.31$  days,  $SD = 33.12$  days, range = 148–269 days) into their pregnancy.

*Personal meaning* was assessed using the Personal Meaning Index from the Life Attitude Profile–Revised (Reker, 2005; Reker & Peacock, 1981). The Life Attitude Profile–Revised is a 48-item measure using a 7-point Likert scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The Personal Meaning Index is composed of two dimensions: purpose (having life goals and a sense of direction, e.g., “I know where my life is going in the future”) and coherence (having a clear and logically integrated sense of self and life in general, e.g., “In thinking of my life, I see a reason for my being here”). Each of these subscales consists of eight items, and responses to all 16 items were summed for a total score. To measure *change in meaning*, we calculated a  $\delta$  score by subtracting prenatal meaning from postnatal meaning for each participant.

*Parenting-related stress* was measured with the Parenting Stress Inventory (Abidin, 1983). The Parenting Stress Index captures parents' perceptions of life stressors caused by the limitations placed upon them due to their parental responsibilities in the first year of their child's life, as well as stressors related to their relationship with their child.

**Table 1**  
*Sample Demographics*

| Variable                                    | Behavioral sample<br><i>N</i> = 88 |           |           | MRI sample<br><i>N</i> = 35 |           |             |
|---------------------------------------------|------------------------------------|-----------|-----------|-----------------------------|-----------|-------------|
|                                             | <i>M</i>                           | <i>SD</i> | Range     | <i>M</i>                    | <i>SD</i> | Range       |
| Prenatal age (years)                        | 32.99                              | 5.65      | 22–57     | 32.70                       | 4.48      | 24–44       |
| Postpartum age (years)                      | 33.49                              | 6.89      | 22–58     | 33.60                       | 4.48      | 24–44       |
| Infant age (months)                         | 6.64                               | 0.63      | 5.47–9.02 | 6.43                        | 0.6       | 5.63–8.2    |
| Days between postpartum visit and scan      |                                    |           |           | 71                          | 114       | 0–436       |
| Prenatal meaning                            | 84.27                              | 16.78     | 49–112    | 86.03                       | 15.68     | 51–111      |
| Postpartum meaning                          | 84.22                              | 16.61     | 48–112    | 87.08                       | 16.50     | 50–112      |
| Parenting stress index                      | 84.52                              | 23.09     | 48–153.33 | 83.11                       | 19.36     | 50.66–124.0 |
| Positive parenting feelings                 | 23.09                              | 6.94      | 10–49     | 36.86                       | 3.08      | 30–40       |
| Negative parenting feelings                 | 36.66                              | 3.63      | 24–40     | 22.92                       | 5.59      | 11–32       |
| Time spent with baby (hr/week)              |                                    |           |           | 41.33                       | 17.57     | 11.5–64     |
| Time as primary caregiver (hr/week)         |                                    |           |           | 19.73                       | 12.19     | 2–46.48     |
| Variable                                    | <i>N</i>                           |           | %         | <i>N</i>                    |           | %           |
| Education                                   |                                    |           |           |                             |           |             |
| High school/General Educational Development | 2                                  |           | 2.27      | 1                           |           | 2.63        |
| Some college                                | 12                                 |           | 13.64     | 6                           |           | 15.79       |
| Associate's degree                          | 2                                  |           | 2.27      | 1                           |           | 2.63        |
| Bachelor's degree                           | 42                                 |           | 47.73     | 12                          |           | 31.58       |
| Master's degree                             | 19                                 |           | 21.59     | 10                          |           | 26.32       |
| Professional or doctoral                    | 11                                 |           | 12.5      | 8                           |           | 21.05       |
| Race/ethnicity                              |                                    |           |           |                             |           |             |
| White                                       | 41                                 |           | 46.59     | 12                          |           | 31.58       |
| Black/African American                      | 6                                  |           | 6.82      | 3                           |           | 7.9         |
| Hispanic or Latino/a                        | 18                                 |           | 20.45     | 12                          |           | 31.58       |
| Asian American and Pacific Islander         | 17                                 |           | 19.32     | 9                           |           | 23.68       |
| Other                                       | 6                                  |           | 6.82      | 2                           |           | 5.26        |

Note. MRI = magnetic resonance imaging.

We additionally used questions from the Parenting Stress Scale to quantify challenges surrounding the parenting experience (Berry & Jones, 1995). Multiple previous studies have used factor analysis to demonstrate that the survey can be adequately split into two subscales representing parental satisfaction (positive parenting feelings) and parental stress (negative parenting feelings; Berry & Jones, 1995; Harding et al., 2020; Pontoppidan et al., 2018). Due to the nature of our hypotheses, we used these two separate scores for exploratory analyses. The items were separated and summed mirroring these previous studies (see Supplemental Materials for the questions in each scale).

Additional covariates (race, ethnicity, income, and education) were collected via self-report at the prenatal visit. The income questionnaire was added halfway through data collection so the sample size for income was  $n = 45$ .

## Behavioral Analyses

All statistical analyses between behavioral variables were performed in RStudio Version 4.1.1. To test associations between key behavioral variables of interest, we performed Pearson's correlations. Since we performed 45 tests, we applied a Holm–Bonferroni to adjust for multiple comparisons. This method sequentially adjusts  $p$  values, starting with the smallest, to control for family-wise error rate. The smallest  $p$  value is compared with  $\alpha/\text{number of tests}$ , and if significant, the next  $p$  value is compared with a progressively less stringent threshold. This process continues until a  $p$  value fails to

meet the adjusted threshold, at which point all subsequent  $p$  values are considered nonsignificant. We applied a Holm–Bonferroni adjusted  $\alpha$  of .05 to determine statistical significance.

## Scan Acquisition

Whole-brain images were collected using a Siemens 3 Tesla MAGNETOM Prisma System scanner using a 20-channel matrix head coil. T1-weighted images were collected using a 3D Magnetization Prepared Rapid Acquisition Gradient Echo (repetition time = 2,530 ms, echo time = 3.13 ms, flip angle = 10°), with an isotropic voxel resolution of 1 mm<sup>3</sup>. Resting state images were collected with T2\* weighted images (210 volumes, repetition time = 2000 ms, echo time = 25 ms, flip angle = 90°, slice thickness = 2.5 mm voxel resolution of 3 × 3 × 2.5 mm). An additional T2 weighted image was taken for review by a neuroradiologist in compliance with university imaging center policies. Prenatal scans were collected 20–36 weeks into pregnancy, generally 1–2 weeks after survey data was collected. Postnatal scans were collected between 6 and 18 months after birth, with this wider range due to the aforementioned data collection disruption from the COVID-19 pandemic in 2020–2021 (mean = 71 days after the postnatal visit, standard deviation = 114).

## Resting-State Preprocessing

Analyses were performed using CONN (Whitfield-Gabrieli & Nieto-Castanon, 2012) release 22.a (Nieto-Castanon & Whitfield-



Gabrieli, 2022) and statistical parametric mapping (Penny et al., 2011) release 12.7771. Functional and anatomical data were preprocessed using a flexible preprocessing pipeline (Nieto-Castanon, 2020) including realignment, outlier detection, indirect segmentation and Montreal Neurological Institute space normalization, and smoothing. Functional data were coregistered to a reference image (first scan of the first session) using a least squares approach and a six parameter (rigid body) transformation without resampling (Friston et al., 1995). Potential outlier scans were identified using Artifact Detection Tools (Whitfield-Gabrieli et al., 2011) as acquisitions with framewise displacement above 0.2 mm or global blood oxygen level-dependent (BOLD) signal changes above 5 SDs (Power et al., 2014). Of the 70 scanning sessions, there was an average of 10.27/210 outlier volumes per session. All but two sessions had more than 150 2 s volumes, with one prenatal session having 135 and one postnatal having 125. We tested whether removing these two would affect our results. A reference BOLD image was computed for each subject by averaging all scans excluding outliers. Functional and anatomical data were coregistered and normalized into standard Montreal Neurological Institute space, segmented into gray matter, white matter, and cerebrospinal fluid tissue classes, and resampled to 2-mm isotropic voxels following an indirect normalization procedure (Calhoun et al., 2017) using statistical parametric mapping unified segmentation and normalization algorithm (Ashburner, 2007; Ashburner & Friston, 2005) with the default Information eXtraction from Images-549 tissue probability map template. Last, functional data were smoothed using spatial convolution with a Gaussian kernel of 8 mm full width half maximum.

## Denoising

In addition, functional data were denoised using a standard denoising pipeline (Nieto-Castanon, 2020) including the regression of potential confounding effects characterized by white matter timeseries (five CompCor noise components), cerebrospinal fluid timeseries (five CompCor noise components), motion parameters and their first order derivatives (12 factors; Friston et al., 1996), and linear trends (two factors) within each functional run, followed by bandpass frequency filtering of the BOLD timeseries (Hallquist et al., 2013) between 0.008 and 0.09 Hz. CompCor noise components within white matter and cerebrospinal fluid were estimated by computing the average BOLD signal as well as the largest principal components orthogonal to the BOLD average, motion parameters within each subject's eroded segmentation masks (Behzadi et al., 2007; Chai et al., 2012). From the number of noise terms included in this denoising strategy, the effective degrees of freedom of the BOLD signal after denoising were estimated to range from 118.9 to 136.9 (average 136.4) across all subjects.

Additionally, all analyses reported were tested excluding the one prenatal and one postnatal scanning sessions that had under 5 min of usable scanning time after scrubbing (4.16 min and 4.5 min, respectively). No significant clusters changed when these data were excluded and only minor changes in total significant voxels occurred.

## First-Level Analysis

Intrinsic connectivity maps characterizing network centrality at each voxel were estimated as the root-mean-square of all short- and long-range connections between a voxel and the rest of the brain

(Martuzzi et al., 2011). Connections were computed from the matrix of bivariate correlation coefficients between the BOLD time series from each pair of voxels, estimated using a singular value decomposition of the  $z$  score normalized BOLD signal (subject-level SVD) with 64 components separately for each subject and condition. Last, intrinsic connectivity measures across voxels were rank sorted and normalized separately for each individual subject and condition using a Gaussian inverse cumulative distribution function with zero mean and unit variance.

## Group-Level Analysis

Group-level analyses were performed using a general linear model (Nieto-Castanon, 2020). For each individual voxel a separate general linear model was estimated, with first-level connectivity measures at this voxel as dependent variables (one independent sample per subject) and groups or other subject-level identifiers as independent variables. Voxel-level hypotheses were evaluated using multivariate parametric statistics with random-effects across subjects and sample covariance estimation across multiple measurements. Inferences were performed at the level of individual clusters (groups of contiguous voxels). Cluster-level inferences were based on parametric statistics from Gaussian random field theory (Worsley et al., 1996). Results were thresholded using a combination of a cluster-forming  $p < .01$  voxel-level threshold and a false discovery rate corrected  $p$ -false discovery rate  $< 0.05$  cluster-size threshold (Chumbley et al., 2010). In each analysis where a postnatal survey measure was associated with resting-state connectivity, we additionally included the number of days between the completion of the questionnaires and the postnatal scan as a covariate, to account for some subjects having longer time between surveys and the scan due to disruptions in data collection caused by the COVID-19 pandemic. Table 2 shows the group-level general

**Table 2**  
*Overview of General Linear Models by Research Question*

| Research question    | Model description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research question 2  | <ul style="list-style-type: none"> <li>Average intrinsic connectivity change from pre- to postnatal, controlling for time between scans</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |
| Research question 3  | <ul style="list-style-type: none"> <li>Prenatal intrinsic connectivity predicted by prenatal personal meaning</li> <li>Postnatal intrinsic connectivity predicted by postpartum meaning, controlling for time between scans</li> <li>Intrinsic connectivity change predicted by postpartum meaning, controlling for time between scans and prenatal meaning</li> </ul>                                                                                                                                                                                        |
| Exploratory analyses | <ul style="list-style-type: none"> <li>Intrinsic connectivity change predicting postpartum meaning, controlling for time between scans, positive parenting feelings, and negative parenting feelings</li> <li>Intrinsic connectivity change predicting positive parenting feelings, controlling for time between scans, postpartum meaning, and negative parenting feelings</li> <li>Intrinsic connectivity change predicting negative parenting feelings, controlling for time between scans, postpartum meaning, and positive parenting feelings</li> </ul> |

linear models we tested for each research question mentioned in the Introduction section.

## Transparency and Openness

This study was not preregistered. The sample size was determined by available grant funding and participant eligibility but also constricted by the COVID-19 pandemic; the initial planned sample was projected to be 10% larger but new data collection was halted in 2020. The sample was drawn from a larger sample of parents who were being followed across the transition to parenthood but we were only able to add the magnetic resonance imaging portion of the study after securing sufficient external grant funding. We specify the conditions under which subjects were excluded. All data and analysis code are available upon reasonable request from the first author.

## Results

### Hypothesis 1: Changes in Personal Meaning

Contrary to our hypothesis, fathers did not report greater levels of meaning after becoming parents ( $t = -0.045, p = .97$ ). Instead, meaning showed relative stability, with prenatal personal meaning significantly predicting meaning at 6 months after birth ( $r = 0.75, p < .001$ ). Of the 88 fathers in our sample, 41 showed decreased levels of meaning, 42 increased, and five remained the same, demonstrating considerable variation in the father's trajectories ( $M_{\text{change}} = -0.57, SD = 11.87$ ; Figure 1). We additionally performed a  $t$  test to assess whether the 10 fathers who completed the personal meaning

survey during the COVID-19 lockdowns differed from those who completed it before, and did not find any significant differences between the two groups ( $M_{\text{pre-pandemic}} = 83.79; M_{\text{pandemic}} = 86.92, p = .56, t = -0.60$ ).

### Hypothesis 2: Associations With Personal Meaning

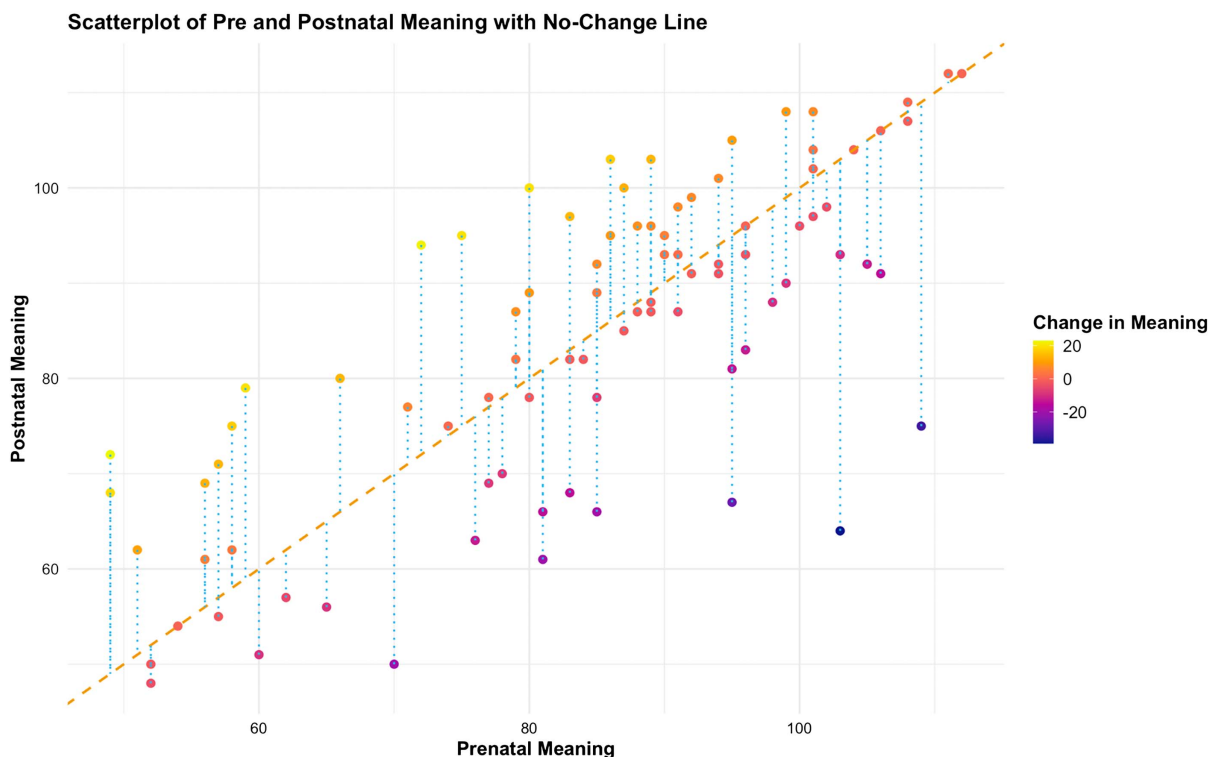
As shown in Table 3, both prenatal and postnatal personal meaning were negatively associated with parenting stress (as measured by the Parental Stress Index). Change in meaning was positively associated with positive parenting feelings. No significant associations were found between personal meaning and negative parenting feelings (as measured by the Parental Stress Scale), age, baby age, education, or household income, either prenatally or postnatally. Of note, the first half of our sample (43 subjects) was missing household income data, resulting in those correlations having less power than the other comparisons. Data were completed for all other associations.

### Hypothesis 3: Intrinsic Functional Connectivity Change During the Transition to Fatherhood

When comparing the prenatal scan to the scan acquired after birth, fathers' intrinsic functional connectivity increased in the right OL, bilateral temporal lobe, right angular gyrus (AG), and thalamus (Th). Intrinsic functional connectivity decreased in the right frontal pole (FP), left opercular cortex, and the anterior cingulate (Figure 2; Table 4).

**Figure 1**

*Change in Personal Meaning Prenatal to Postnatal*



*Note.* Each point represents a subject's prenatal (x axis) and postnatal (y axis) levels of personal meaning. The dotted line represents no change in personal meaning, with points above and below it being increases and decreases, respectively. See the online article for the color version of this figure.

**Table 3**  
*Pearson Correlations Among Personal Meaning and Parenting Outcomes*

| Variable                             | 1      | 2       | 3    | 4       | 5       | 6    | 7    | 8    | 9   | 10 |
|--------------------------------------|--------|---------|------|---------|---------|------|------|------|-----|----|
| 1. Prenatal meaning                  | —      |         |      |         |         |      |      |      |     |    |
| 2. Postpartum meaning                | .75*** | —       |      |         |         |      |      |      |     |    |
| 3. Change in meaning                 | -.38** | .34     | —    |         |         |      |      |      |     |    |
| 4. Parenting stress (PSI)            | -.35*  | -.45*** | -.14 | —       |         |      |      |      |     |    |
| 5. Positive parenting feelings (PSS) | .07    | .33     | .37* | -.65*** | —       |      |      |      |     |    |
| 6. Negative parenting feelings (PSS) | -.27   | -.33    | -.10 | .71***  | -.49*** | —    |      |      |     |    |
| 7. Age at postpartum visit (years)   | -.04   | .03     | .10  | -.19    | .07     | -.08 | —    |      |     |    |
| 8. Baby age (months)                 | -.2    | -.18    | .03  | .16     | -.004   | .10  | -.10 | —    |     |    |
| 9. Education                         | .04    | -.05    | -.12 | -.07    | -.03    | .02  | .15  | .10  | —   |    |
| 10. Household income                 | -.001  | -.07    | -.11 | -.002   | -.10    | -.02 | .24  | -.04 | .38 | —  |

*Note.* An initial  $\alpha$  of .05 is selected and then Holm–Bonferonni correction is applied. PSI = Parental Stress Index; PSS = Parental Stress Scale.  
\*  $p < .05$ . \*\*  $p < .01$ . \*\*\*  $p < .001$  after Holm–Bonferonni correction.

**Hypothesis 4: Intrinsic Functional Connectivity Association With Personal Meaning**

Prenatal resting-state connectivity was not significantly associated with prenatal levels of meaning. Postnatal resting-state connectivity was associated with postnatal meaning. Specifically, greater postnatal connectivity in bilateral inferior frontal gyrus predicted higher postnatal meaning whereas connectivity in the right cerebellum, and left AG was negatively associated with postnatal meaning (Figure S1; Table 4).

Change in intrinsic connectivity was also associated with postnatal meaning. Greater postnatal connectivity in the right insular cortex, right temporal pole (TP), right middle temporal gyrus, left parahippocampal gyrus, and right hippocampus predicted higher postnatal meaning. In contrast, postnatal connectivity in the left cerebellum, left lateral occipital cortex, and left caudate predicted lower postnatal meaning (Figure 3; Table 4).

**Exploratory Hypothesis: Associations Between Meaning and Positive and Negative Parental Feelings With Change in Intrinsic Functional Connectivity**

When controlling for time between scans and both positive and negative parenting feelings, the change in intrinsic connectivity was

positively associated with postnatal levels of meaning in the Th, right Heschl’s gyrus, right insular cortex, and left FP. There was a negative association with a large swath of lateral occipital lobe (OL). Positive parenting feelings (when controlling for the other two measures and time from survey to scan) was positively associated with change in the connectivity of the middle frontal gyrus and the left supramarginal gyrus. There were negative associations with the caudate, right cerebellum, and the anterior cingulate. Finally, negative parenting feelings had independent positive associations with intrinsic connectivity change in the left postcentral gyrus and bilateral cerebellum (Table 4, Figure 4).

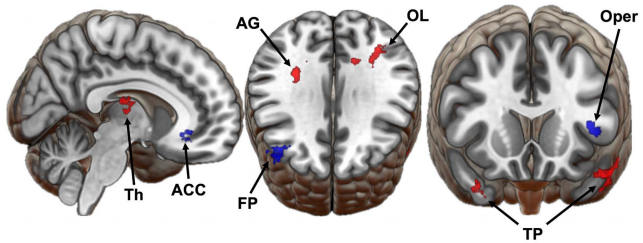
**Discussion**

We found that first-time fathers showed largely stable levels of self-reported personal meaning across the transition to parenthood, with some fathers showing an increased sense of meaning and other fathers showing a decreased sense of meaning. Personal meaning was negatively associated with parenting stress and negative feelings about parenting and positively associated with positive feelings about parenting. When comparing the prenatal scan with the scan acquired after birth, we observed significant changes in men’s resting-state functional connectivity, with increased connectivity in regions such as the bilateral TPs, right AG, Th, and right lateral occipital cortex, and decreased connectivity in the right FP, left opercular cortex, and anterior cingulate. Notably, increased connectivity in the insular cortex and TP was associated with greater endorsement of postnatal meaning, suggesting that neural plasticity in these regions specifically may support fathers’ psychological adaptation to parenthood. Our results reaffirm that the transition to fatherhood can involve significant psychobiological changes that vary across individuals.

**Change in Meaning**

Surprisingly, we did not find that fathers on average reported greater meaning over the transition to fatherhood. While some studies report that parents generally experience higher levels of meaning and well-being than nonparents, other work has found that the relationship between parenting and meaning is complex and variable (Nelson et al., 2014; Pancer et al., 2000). The first year of parenting is a particularly fraught transition period, and qualitative studies of

**Figure 2**  
*Intrinsic Resting-State Functional Connectivity Change During the Transition to Fatherhood*



*Note.* Red = clusters with significant increases in intrinsic functional connectivity; Blue = clusters with significant decreases in intrinsic connectivity; Th = thalamus; ACC = anterior cingulate cortex; AG = angular gyrus; FP = frontal pole; OL = lateral occipital lobe; Oper = operculum; TP = temporal pole. See the online article for the color version of this figure.

**Table 4***Clusters and Peak Coordinates From Intrinsic Connectivity Analyses*

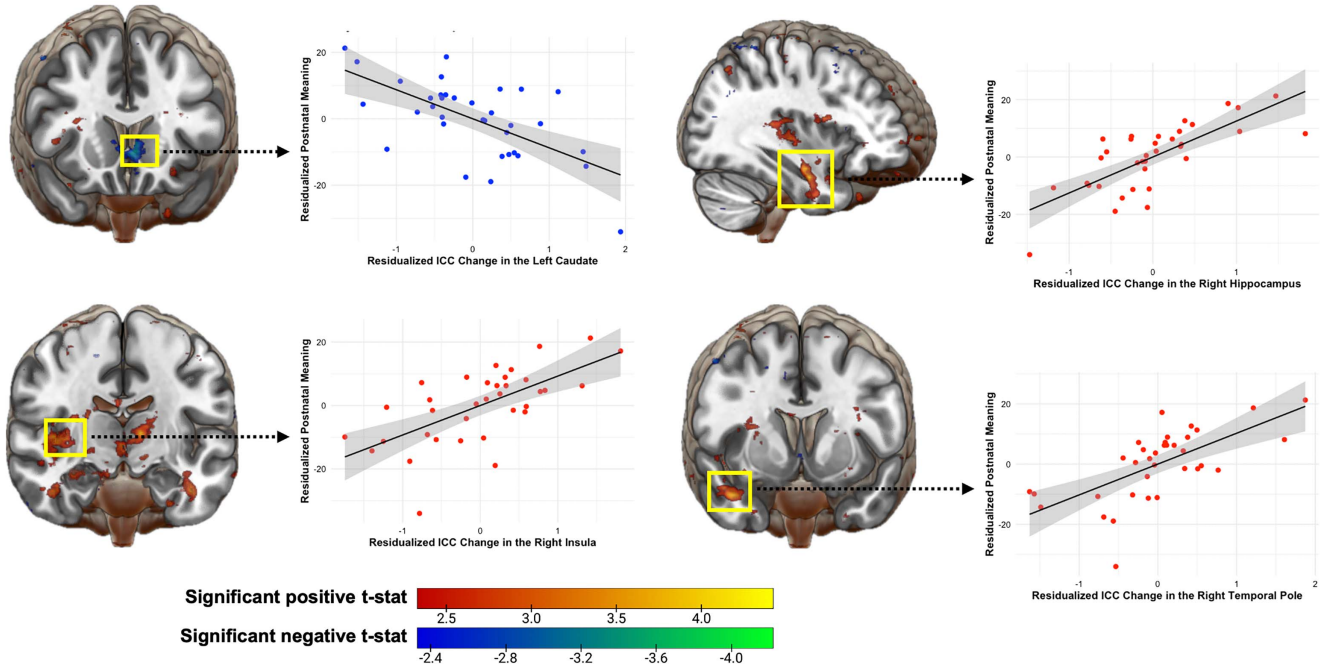
| Region                                                                                                                                   | Peak coordinate (MNI: <i>x, y, z</i> ) | <i>K</i> (# voxel) | Connectivity change | Peak <i>t</i> value |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------|---------------------|---------------------|
| Hypothesis 3: Intrinsic functional connectivity change during the transition to fatherhood                                               |                                        |                    |                     |                     |
| Average intrinsic connectivity change (post > pre)                                                                                       |                                        |                    |                     |                     |
| Right lateral occipital lobe                                                                                                             | (-18, -48, 20)                         | 168                | Increased           | 4.9                 |
| Left temporal lobe                                                                                                                       | (-48, 14, -34)                         | 136                | Increased           | 4.2                 |
| Right angular gyrus                                                                                                                      | (32, -38, 26)                          | 125                | Increased           | 4.7                 |
| Thalamus                                                                                                                                 | (-6, -10, 14)                          | 112                | Increased           | 4.7                 |
| Right temporal lobe                                                                                                                      | (38, 20, -34)                          | 45                 | Increased           | 4.4                 |
| Right frontal pole                                                                                                                       | (36, 30, 24)                           | 106                | Decreased           | -3.9                |
| Left opercular cortex                                                                                                                    | (-40, 8, 6)                            | 55                 | Decreased           | -3.6                |
| Anterior cingulate                                                                                                                       | (-2, 28, -6)                           | 47                 | Decreased           | -4.9                |
| Hypothesis 4: Intrinsic functional connectivity association with personal meaning                                                        |                                        |                    |                     |                     |
| Prenatal connectivity ~ prenatal personal meaning<br>(No significant results)                                                            |                                        |                    |                     |                     |
| Postnatal connectivity ~ postnatal meaning                                                                                               |                                        |                    |                     |                     |
| Left inferior frontal gyrus                                                                                                              | (-56, 12, 12)                          | 114                | Increased           | 4.2                 |
| Right inferior frontal gyrus                                                                                                             | (46, 14, 4)                            | 81                 | Increased           | 4.8                 |
| Left angular gyrus                                                                                                                       | (40, -48, -32)                         | 138                | Decreased           | -6.0                |
| Right cerebellum                                                                                                                         | (-40, -48, -32)                        | 73                 | Decreased           | -5.5                |
| Connectivity change (post > pre) ~ postnatal meaning<br>(controlling for prenatal meaning)                                               |                                        |                    |                     |                     |
| Right hippocampus                                                                                                                        | (34, -6, -24)                          | 82                 | Increased           | 4.8                 |
| Right insular cortex                                                                                                                     | (42, -16, 4)                           | 73                 | Increased           | 4.3                 |
| Right temporal pole                                                                                                                      | (40, 6, -26)                           | 68                 | Increased           | 4.6                 |
| Left parahippocampal gyrus                                                                                                               | (-38, -6, -28)                         | 67                 | Increased           | 4.6                 |
| Right middle temporal gyrus                                                                                                              | (-40, -34, -8)                         | 64                 | Increased           | 4.8                 |
| Left cerebellum                                                                                                                          | (-42, -56, -28)                        | 109                | Decreased           | -4.4                |
| Left lateral occipital cortex                                                                                                            | (-46, -62, 32)                         | 68                 | Decreased           | -4.4                |
| Left caudate                                                                                                                             | (-4, 12, 4)                            | 65                 | Decreased           | -4.2                |
| Exploratory analyses                                                                                                                     |                                        |                    |                     |                     |
| Connectivity change (post > pre) ~ postnatal meaning<br>(controlling for positive + negative parenting feelings)                         |                                        |                    |                     |                     |
| Thalamus                                                                                                                                 | (2, -16, -10)                          | 102                | Increased           | 4.3                 |
| Right Heschl's gyrus                                                                                                                     | (38, -24, 6)                           | 92                 | Increased           | 4.5                 |
| Right insular cortex                                                                                                                     | (34, 6, 16)                            | 75                 | Increased           | 4.5                 |
| Left frontal pole                                                                                                                        | (-14, 46, -20)                         | 70                 | Increased           | 4.7                 |
| Lateral occipital lobe                                                                                                                   | (-30, -78, 22)                         | 571                | Decreased           | -5.3                |
| Connectivity change (post > pre) ~ positive parenting<br>feelings (controlling for negative parenting feelings<br>and postnatal meaning) |                                        |                    |                     |                     |
| Left supramarginal gyrus                                                                                                                 | (-58, -32, 20)                         | 117                | Increased           | 4.9                 |
| Middle frontal gyrus                                                                                                                     | (-50, 26, 26)                          | 82                 | Increased           | 4.5                 |
| Caudate                                                                                                                                  | (4, 14, -4)                            | 180                | Decreased           | -5.4                |
| Right cerebellum                                                                                                                         | (32, -66, -36)                         | 76                 | Decreased           | -4.6                |
| Anterior cingulate                                                                                                                       | (-8, 36, 6)                            | 78                 | Decreased           | -5.2                |
| Connectivity change (post > pre) ~ negative parenting<br>feelings (controlling for positive parenting feelings<br>and postnatal meaning) |                                        |                    |                     |                     |
| Left postcentral gyrus                                                                                                                   | (-36, -28, 54)                         | 142                | Increased           | 4.3                 |
| Left cerebellum                                                                                                                          | (-38, -44, -52)                        | 79                 | Increased           | 4.8                 |
| Right cerebellum                                                                                                                         | (2, -84, -28)                          | 76                 | Increased           | 5.1                 |

*Note.* Peak coordinates and *t* values from all intrinsic connectivity analyses grouped by hypothesis. All analyses involving change in connectivity from prenatal to postnatal time points additionally controlled for time between postnatal behavioral data collection and postnatal scan. MNI = Montreal Neurological Institute.

meaning during this timeframe often include more reports of the challenging nature of shifting identities and expectations (Höfner et al., 2011; Pancer et al., 2000; Prinds et al., 2014). In this way, meaning may be a particularly strong marker of parental adaptation, and one that is not associated with sociodemographic factors such as education or income. For example, fathers who reported higher levels of meaning during this transition also reported lower levels of parenting stress. These results converge with prior studies showing

that personal meaning is associated with reduced mental health risks during transformative, yet stressful, life stages (Boreham & Schutte, 2023; Dulaney et al., 2018; Hill et al., 2018; Krause, 2007). Higher postnatal personal meaning may reflect integration of the fatherhood role into one's self-concept, providing some sense of control against challenges during this new stressful period (Garfield et al., 2014; Genesoni & Tallandini, 2009; Strauss & Goldberg, 1999).



**Figure 3***Change in Intrinsic Functional Connectivity Predictive of Postnatal Meaning*

*Note.* *T* stat map of intrinsic connectivity change associated with postnatal levels of meaning, and scatterplots showing the strength of the effect in key significant clusters. See the online article for the color version of this figure.

### Changes in Intrinsic Functional Connectivity

We found significant increases and decreases in intrinsic connectivity across several brain regions during the transition to fatherhood. Notably, increases in connectivity were observed in areas such as the bilateral temporal lobes, right AG, Th, and right lateral occipital cortex, while decreases were found in the right FP, left opercular cortex, and anterior cingulate cortex. The observed increases in connectivity in regions such as the temporal lobes and AG may reflect enhanced processing of socially relevant and emotionally salient stimuli, consistent with prior studies linking these areas to caregiving behaviors (Dufford et al., 2019; Orchard et al., 2023). These regions have been implicated in mentalizing, empathy, and autobiographical memory—all processes that may be critical for forming and sustaining relationships with one's child (Hoekzema et al., 2022; Orchard et al., 2023). In contrast, decreases in connectivity within the anterior cingulate and FP may reflect a pruning or streamlining of neural networks as fathers adapt to the cognitive and emotional demands of parenting (Martínez-García et al., 2023; Saxbe & Martínez-García, 2024). These findings provide initial evidence of functional brain changes, complementing the existing literature on structural reorganization during the transition to fatherhood.

### Connectivity Changes Related to Meaning

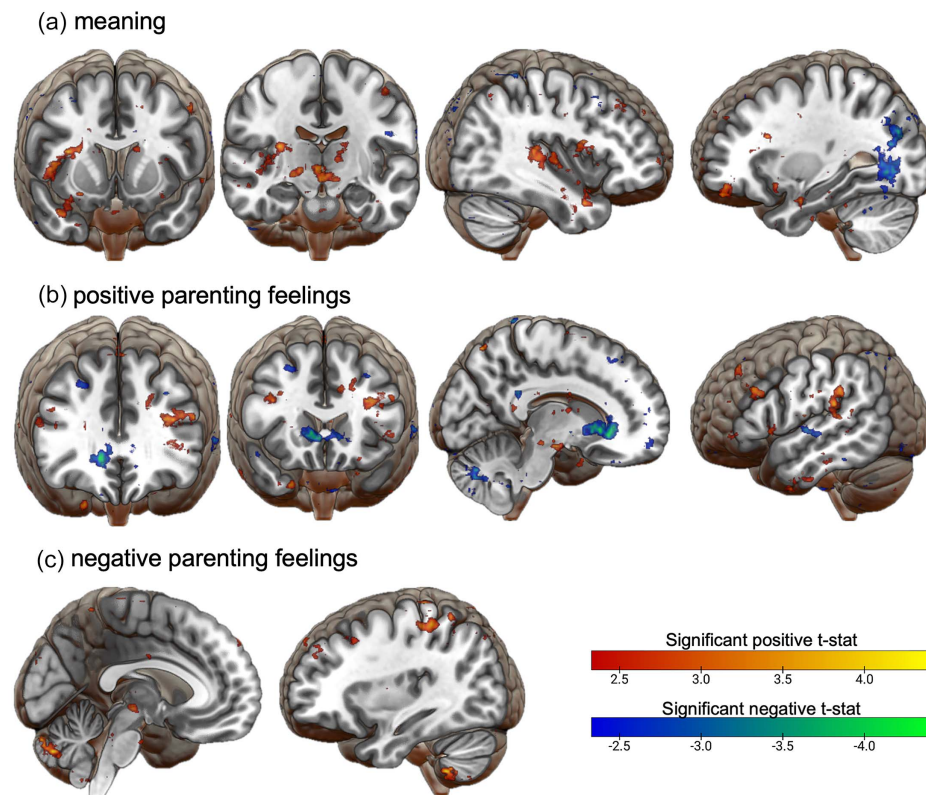
Based on previous literature, we hypothesized that both current resting-state connectivity, and functional changes from prenatal to 6 months postnatal in the insular cortex and DMN, would predict

higher levels of postnatal meaning. Contrary to our expectations, postnatal personal meaning was not associated cross-sectionally with resting-state connectivity in these regions, instead being associated with the inferior frontal gyri, AG, and cerebellum. However, changes in connectivity of the TP and broader temporal gyrus, insula, parahippocampus, and hippocampus across the transition to fatherhood were associated with greater postnatal meaning. These regions have frequently been implicated in functional studies of parenting as important for the mentalizing and reward-related processes involved in caretaking (Hoekzema et al., 2022; Horstman et al., 2022; Orchard et al., 2023). The insula's relationship with meaning is especially interesting in the context of the transition to parenthood, as previous work has found that its activity when viewing infant caregiving stimuli can successfully distinguish between expectant fathers and other men (Díaz-Rojas et al., 2021). As these men are not yet involved in caregiving, yet still are distinguishable based on this insular activity, these prior results may suggest that these changes begin early from psychological preparation to become a father. This psychological preparation is possibly related to recontextualizing one's personal meaning and purpose.

Conversely, meaning was associated with decreases in connectivity in the caudate, cerebellum, and OL. Speculatively, decreases in the overall connectivity of these regions, while other regions increase, may reflect an increase in the selectivity in connections—that is, adaptation by strengthening a few functional connections while most become weaker. The differences between regions associated with connectivity at a single time point versus those that changed over time suggest that maintaining meaning after a child's

**Figure 4**

*Intrinsic Functional Connectivity Change Associated With Postnatal Meaning and Parenting Feelings*



*Note.* Change in intrinsic connectivity pre- to postnatal predicted postnatal levels of personal meaning, positive parenting feelings, and negative parenting feelings, while controlling for the other two variables and scanning delays. See the online article for the color version of this figure.

arrival relies on dynamic brain adjustments rather than static connectivity patterns. This highlights the importance of neural adaptation in responding to the challenges of new parenthood. Consequently, it is the changes in connectivity that are more strongly linked to maintaining higher levels of meaning postnatally.

The change in connectivity of the TP was strongly related with fathers' sense of meaning. The TP plays a role in a variety of higher cognitive functions, many of which involve associating visceral affective information with the meaning of sensory inputs (Olson et al., 2007; Wong & Gallate, 2012). This processing is important for affective theory of mind, especially when decoding the emotional states of an individual, which requires prior knowledge or familiarity of that person (Caillaud et al., 2020; Mier et al., 2010). Previous research has found that the TPs are more active when parents view interactions with their own infants (Atzil et al., 2012) or listen to infant cries (Khoddam et al., 2020; Witteman et al., 2019). These results may suggest a role for the TP's functional relevance to affective theory of mind in parenting. Interestingly, the TPs of older men who are fathers have been found to have greater gray matter density than in those who are not fathers (Orchard et al., 2020). It is possible that the resting-state changes we found reflect ongoing functional development through interaction with the newborn.

### Meaning, Positive, and Negative Parenting Feelings as Distinct Contributors to Neural Change

Our exploratory analyses further dissected the unique contributions of meaning, parental satisfaction, and parenting stressors to changes in intrinsic connectivity during the transition to fatherhood. Each analysis controlled for the other variables to isolate the neural effects specific to each construct. For meaning, increased connectivity was observed in the Th, right Heschl's gyrus, right insular cortex, and left FP, while decreased connectivity was found in the OL and left caudate. In contrast, positive parenting feelings were associated with positive connectivity changes in the middle frontal gyrus and left supramarginal gyrus, and negative changes in the anterior cingulate and caudate. Parenting stressors, on the other hand, showed distinct positive associations with connectivity in the left postcentral gyrus and bilateral cerebellum.

Early parenthood is characterized by a mix of positive and negative emotions, as many new parents experience both joy and stress simultaneously (Baldwin et al., 2018; Blakey et al., 2025; Meleagrou-Hitchens & Willig, 2022; Solberg et al., 2023). The findings from this study suggest that the neural effects of meaning-making during this transition extend beyond a simple dichotomy of

positive versus negative feelings. The connectivity changes linked to meaning suggest that it may play a broader integrative role, supporting fathers in the integration of novel mixed affective experiences into a coherent narrative. This ability to contextualize and reframe emotional experiences may provide resilience despite the day-to-day affective climate (Hill et al., 2018; Shikimoto et al., 2021). By contrast, the patterns associated with positive and negative parenting feelings may reflect more immediate, affective responses to the parenting experience.

## Future Directions

Our study shaded unique light on how personal meaning is associated with parenting stress, affective experiences, and intrinsic connectivity in the transition to fatherhood. However, our study had several limitations. Our study was specific to cohabiting heterosexual fathers in the greater Los Angeles area, which may limit the generalizability of our results. Relatedly, a key limitation of our study is the absence of a control group of nonparents. This limits our ability to determine whether the observed neural changes and associations with personal meaning are specific to the transition to fatherhood or reflect how men adapt psychobiologically to major life changes more broadly.

Furthermore, our sample size is limited to 35 subjects who had usable data at both timepoints, and some of these scans were delayed due to data collection interruptions during the COVID-19 pandemic lockdowns. The COVID-19 pandemic may have introduced unique stressors and dynamics surrounding parenting that complicate comparisons with nonpandemic contexts (Taubman-Ben-Ari et al., 2021). However, we found no significant differences in levels of meaning between fathers tested before versus during the pandemic, suggesting our results are not solely influenced by pandemic exposure. Despite these limitations, this work shaded preliminary light on how meaning tracks with the transition to parenthood. Future work can investigate the broad applicability of these findings by researching these relationships in mothers, adoptive parents, and nonresidential fathers. Additionally, meaning in life may serve as a potential basis for intervention in parents-to-be, and clinical studies may aim to investigate whether these associations hold when sense of meaning is directly supported through therapeutic interventions. Interventions that incorporate an emphasis on personal meaning could be adapted to support new parents (Byrne et al., 2021; Whittingham et al., 2016). Finally, a more detailed understanding of why subjects respond to the personal meaning survey in the manner they do would help shed light on many of the potential interpretations of the data. It is possible that fathers report higher or lower levels of meaning for different reasons during the prenatal and postnatal period. Different sources of finding meaning during these different time periods may also explain differences in the neurobiological correlates. Future research could incorporate qualitative interviewing to further understand these constructs.

In conclusion, fatherhood may have variable psychological and neurobiological effects, with some of this variability being reflected by their broader sense of meaning. Our study suggested that men's subjective feelings about the meaning and purpose of their lives can be an informative marker of these seen and unseen changes. These neural changes may be involved in successful adaptation to parenthood. Increased connectivity in regions such as the TPs and insular cortex may aid in reappraising, contextualizing, and emotionally

engaging with the demands of parenthood. In this way, personal meaning may not only index psychological well-being but also align with specific neural pathways that support the adaptation to the new functional role of parenting.

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